

NLP Introduction & Text Processing Assignment

Question 1

What is Computational Linguistics and how does it relate to NLP?

Answer:

Computational Linguistics is the study of language using computational methods and algorithms. It focuses on modeling human language using computers.

Relation to NLP:

- Computational Linguistics provides the **theoretical foundation**
- NLP is the **practical implementation** of those concepts

Question 2

Briefly describe the historical evolution of Natural Language Processing.

Answer:

Answer:

1. **Chatbots & Virtual Assistants (e.g., Alexa, Siri)**
2. **Sentiment Analysis (analyzing opinions from reviews)**
3. **Machine Translation (Google Translate)**

Question 3

List and explain three major use cases of NLP in today's tech industry.

Answer:

- **1950s–60s:** Rule-based systems
- **1980s–90s:** Statistical methods introduced
- **2000s:** Machine Learning models
- **2010s–Present:** Deep Learning & Transformers (e.g., BERT, GPT)

Question 4

What is text normalization and why is it essential in text processing tasks?

Answer:

Text normalization is the process of converting text into a standard format.

Examples:

- Lowercasing
- Removing punctuation
- Removing stopwords

Importance:

- Improves model accuracy
- Reduces noise
- Makes processing easier

Question 5

Compare and contrast stemming and lemmatization with suitable examples.

Answer:

Feature	Stemming	Lemmatization
Method	Cuts words	Uses dictionary
Output	Not always valid	Meaningful words
Example	running → run	better → good

Question 6

Write a Python program that uses regular expressions (regex) to extract all email addresses from the given block of text.

Answer:

```
import re

text = """Hello team, please contact us at support@xyz.com for technical issues,
or reach out to our HR at hr@xyz.com. You can also connect with John at
john.doe@xyz.org and jenny via jenny_clarke126@mail.co.us.
For partnership inquiries, email partners@xyz.biz."""

emails = re.findall(r'[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}', text)

print(emails)
```

Output:

```
['support@xyz.com', 'hr@xyz.com', 'john.doe@xyz.org', 'jenny_clarke126@mail.co.us',
'partners@xyz.biz']
```

Question 7

Given the sample paragraph, perform string tokenization and frequency distribution using Python and NLTK.

Answer:

```

import nltk
from nltk.tokenize import word_tokenize
from nltk.probability import FreqDist

nltk.download('punkt_tab') # Changed to download 'punkt_tab' as per error message
nltk.download('punkt') # Keep 'punkt' download as it is the standard tokenizer

text = """Natural Language Processing (NLP) is a fascinating field that combines linguistics,
computer science, and artificial intelligence."""

tokens = word_tokenize(text.lower())

fdist = FreqDist(tokens)

print(fdist.most_common(10))

```

Output:

```

[nltk_data] Downloading package punkt_tab to /root/nltk_data...
[nltk_data]   Unzipping tokenizers/punkt_tab.zip.
[nltk_data] Downloading package punkt to /root/nltk_data...
[(',', 2), ('natural', 1), ('language', 1), ('processing', 1), ('(', 1), ('nlp', 1),
(')', 1), ('is', 1), ('a', 1), ('fascinating', 1)]
[nltk_data]   Package punkt is already up-to-date!

```

Question 8

Create a custom annotator using spaCy or NLTK that identifies and labels proper nouns in a given text.

```

import spacy

nlp = spacy.load("en_core_web_sm")

text = "John lives in New York and works at Google."

doc = nlp(text)

for token in doc:

    if token.pos_ == "PROPN":

        print(token.text, "-> Proper Noun")

```

Output:

```

John -> Proper Noun
New -> Proper Noun
York -> Proper Noun
Google -> Proper Noun

```

Question 9

Using Gensim, demonstrate how to train a simple Word2Vec model on the given dataset. Tokenize, preprocess, and train the model.

Answer:

```
from gensim.models import Word2Vec
from nltk.tokenize import word_tokenize

dataset = [
    "Natural language processing enables computers to understand human language",
    "Word embeddings are a type of word representation",
    "Word2Vec is a popular technique",
    "Text preprocessing is important",
    "Tokenization helps clean text"
]

# Tokenization
data = [word_tokenize(sentence.lower()) for sentence in dataset]

# Train model
model = Word2Vec(sentences=data, vector_size=100, window=5, min_count=1, workers=4)

# Example
print(model.wv['language'])
```

Output

```
[[-0.00713902  0.00124103 -0.00717672 -0.00224462  0.0037193  0.00583312
  0.00119818  0.00210273 -0.00411039  0.00722533 -0.00630704  0.00464722
 -0.00821997  0.00203647 -0.00497705 -0.00424769 -0.00310898  0.00565521
  0.0057984 -0.00497465  0.00077333 -0.00849578  0.00780981  0.00925729
 -0.00274233  0.00080022  0.00074665  0.00547788 -0.00860608  0.00058446
  0.00686942  0.00223159  0.00112468 -0.00932216  0.00848237 -0.00626413
 -0.00299237  0.00349379 -0.00077263  0.00141129  0.00178199 -0.0068289
 -0.00972481  0.00904058  0.00619805 -0.00691293  0.00340348  0.00020606
  0.00475375 -0.00711994  0.00402695  0.00434743  0.00995737 -0.00447374
 -0.00138926 -0.00731732 -0.00969783 -0.00908026 -0.00102275 -0.00650329
  0.00484973 -0.00616403  0.00251919  0.00073944 -0.00339215 -0.00097922
  0.00997913  0.00914589 -0.00446183  0.00908303 -0.00564176  0.00593092
 -0.00309722  0.00343175  0.00301723  0.00690046 -0.00237388  0.00877504
  0.00758943 -0.00954765 -0.00800821 -0.0076379  0.00292326 -0.00279472
 -0.00692952 -0.00812826  0.00830918  0.00199049 -0.00932802 -0.00479272
  0.00313674 -0.00471321  0.00528084 -0.00423344  0.0026418 -0.00804569
  0.00620989  0.00481889  0.00078719  0.00301345]
```

Question 10

Imagine you are a data scientist at a fintech startup. Outline the steps to clean, process, and extract useful insights from customer reviews using NLP techniques.

Answer:**Steps:**

1. **Data Collection**
 - Gather customer reviews
2. **Data Cleaning**
 - Remove noise, punctuation, stopwords
3. **Text Processing**
 - Tokenization

- Lemmatization
- 4. **Feature Extraction**
 - TF-IDF / Word embeddings
- 5. **Model Building**
 - Sentiment analysis model (e.g., Logistic Regression / Neural Network)
- 6. **Handling Large Data**
 - Batch processing
 - Distributed computing
- 7. **Evaluation**
 - Precision, Recall, F1-score
- 8. **Insights**
 - Identify common complaints
 - Detect positive/negative trends